Predictive Grid Management for Heat Pump Networks

Background & Context

The mass electrification of heating in Poland is a cornerstone of the energy transition. However, for Distribution System Operators (DSOs), the rapid influx of heat pumps creates significant challenges. Precise forecasting of aggregated power demand is essential to prevent grid overloads and ensure system stability. This challenge focuses on high-resolution data analysis to predict seasonal grid strain.

Problem Statement

The goal of this challenge is to develop a high-precision predictive system that utilizes historical data from 2024 to forecast the total electrical load for the summer–autumn of 2025. This is an extrapolation task; participants must build models capable of projecting learned behaviors into a new year.

- Timeframe: Forecast May-October 2025 (6 months per device).
- Task Nature: This is an extrapolative challenge. Models must be designed to project demand beyond the provided historical training scope, accounting for shifting patterns in device behavior.
- Objective: Architect a system that generates precise per-device monthly load forecasts.

Data Specification

The dataset provided is of high resolution, requiring efficient data handling:

- Frequency: Data is recorded at 5-minute intervals.
- Data Management: Due to the substantial volume of data, teams are expected to "slice" or subsample the dataset for efficient training and model development.
- No Synthesis: The dataset consists of real-world telemetry; there is no requirement for generating additional synthetic data.

Download ensemble_task_3_data_oct24-oct25.tar.gz

Dataset Splits

Subset	Period	Months	Notes
Train	Oct 2024 – Apr 2025	7 months	Full feature set
Validation	May 2025 – Jun 2025	2 months	Partial feature set; x2 withheld

Test	Jul 2025 – Oct 2025	4 months	Partial feature set; x2 withheld

Feature Reference

data.csv – time-series telemetry, one row per device per 5-minute interval:

Column	Feature	Description
deviceId	Device ID	Unique identifier for each unit.
Timedate	Timestamp	UTC timestamp of the 5-minute reading.
t1	External temperature	Outdoor ambient temperature; min-max normalised.
t2	Internal temperature	Indoor ambient temperature; min-max normalised.
t3	Source HEX temperature 1	Temperature 1 on the source-side heat exchanger; min-max normalised.
t4	Source HEX temperature 2	Temperature 2 on the source-side heat exchanger; min-max normalised.
t5	Load HEX temperature 1	Temperature 1 on the receive-side heat exchanger; min-max normalised.
t6	Load HEX temperature 2	Temperature 2 on the receive-side heat exchanger; min-max normalised.
t7	DHW tank temperature	Domestic hot water storage tank temperature; min-max normalised.
t8	Cooling circuit temperature 1	Temperature 1 of the cooling circuit; min-max normalised.
t9	CH buffer temperature	Central heating buffer tank temperature; min-max normalised.
t10	Source HEX air temperature 1	Air temperature 1 at the source-side heat exchanger; min-max normalised.
t11	Source HEX air temperature 2	Air temperature 2 at the source-side heat exchanger; min-max normalised.
t12	Load HEX temperature 3	Temperature 3 on the receiver-side heat exchanger; min-max normalised.
t13	Load HEX temperature 4	Temperature 4 on the receiver-side heat exchanger; min-max normalised.
x1	Operating frequency	Compressor/device operating frequency.
x2	Grid load indicator	Individual electrical grid load index *(prediction target)*.
x3	Heating curve type	Type of heating curve configuration used by the device.
deviceType	Device type	Integer-encoded device category.

devices.csv – static per-device metadata:

Column	Feature	Descripton
deviceId	Device ID	Unique identifier for each unit (join key).
latitude	Latitude	Geographic latitude of the device installation site.
longitude	Longitude	Geographic longitude of the device installation site.

Prediction Target

For each device d and forecast month m, predict the **average x2** over all 5-minute readings within that month:

$$target_{d,m} = \frac{1}{N_{d,m}} \sum_{i=1}^{N_{d,m}} x_2^{(d,m,i)}$$

where $N_{d,m}$ is the total number of readings for device d in month m.

Submission Format

Submit a single CSV file covering **both validation and test months** — one row per device per month:

```
deviceId,year,month,prediction
000cc3cb...,2025,5,0.18908
000cc3cb...,2025,6,0.18908
000cc3cb...,2025,7,0.21034
...
```

Column	Type	Description
deviceId	str	Device identifier (matches training data)
year	int	Year of the forecast month
Month	int	Month number (5–10 for May–October 2025)
prediction	float	Predicted average x2 for that device and month

Evaluation Metrics

Submissions are evaluated using MAE (Mean Absolute Error):

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

where n is the total number of predictions. Lower values indicate better predictions, and a score of 0 corresponds to a perfect prediction.

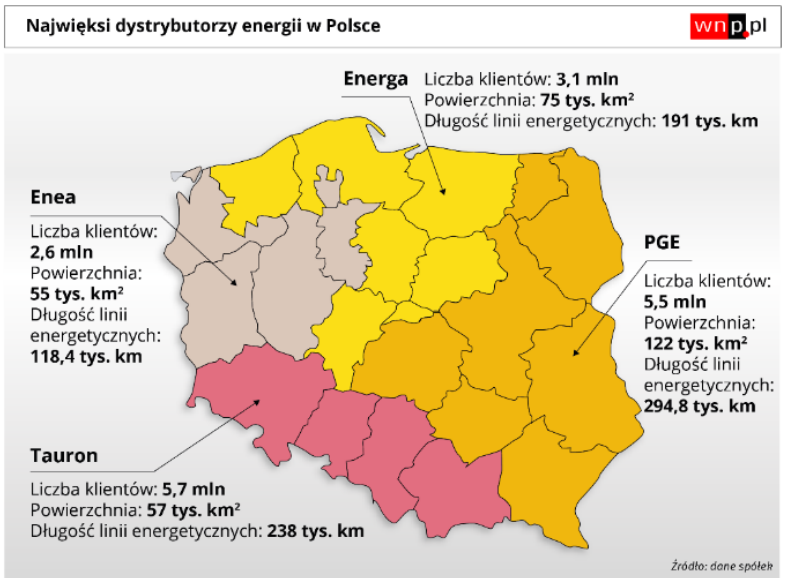
Leaderboard vs. Final Score

Score	Months used	Weights
Leaderboard (visible)	Validation only (May – Jun 2025)	-
Final score	Validation + Test (May – Oct 2025)	2/6 valid 4/6 test

In order to be included in the final leaderboard, a reproducible code implementation for generating the submission should be provided, in a form of a link to public GitHub repository.

Additional Information

The map below presents the regional coverage areas of the four largest electricity distribution companies in Poland: Enea, Energa, PGE, and Tauron.



Source: wnp.pl